

Zen PDF Studio & GDRM Ecosystem

Comprehensive User Guide, Architecture & Technical Manual

Executive Overview:

Zen PDF Studio is a modern, ultra-fast, and privacy-first universal document application for Android and Windows. It provides native in-app editing studios for Office documents (Word, Excel, PowerPoint), code, and HTML, combined with enterprise-grade **GDRM (Granular Digital Right Manager) Zero-Trust** encryption, Aadhaar/PAN cryptographic signature verification, and instant zero-latency photo-to-PDF compilation.

1. GDRM Zero-Trust Encryption & Cryptographic Container (.gdrm)

Zen PDF integrates the **GDRM (Granular Digital Right Manager)** ecosystem (powered by `gdrm_sdk`), allowing users to lock PDF documents into military-grade encrypted `.gdrm` packages with hardware and biometric-level security policies:

- **Machine Hardware Binding (K_{device}):** Cryptographically locks document decryption exclusively to the authoring machine or a specified target device ID. If forwarded to another PC or phone, the container remains mathematically impenetrable.
- **ChronoLock Time Bomb Expiry:** Configure auto-expiring documents (1 hour, 24 hours, 3 days, 7 days, or precise UTC timestamps). After expiration, access is permanently refused.
- **Target User Locking:** Seal sensitive materials to a designated `@username` identity.
- **Indelible Forensic Watermark:** Overlays dynamic, non-removable diagonal watermarks with host hardware fingerprint, IP, and timestamp across all rendered pages.
- **Physical Print & Clipboard Restrictions:** Hard-disables OS text extraction, screenshots, and unauthorized physical printing.
- **SnailTrail Immutable Audit Logs:** Embedded cryptographic ledger tracking authoring timestamp, device fingerprint, and tampering verification.

2. Universal In-App Document Studios

Zen PDF goes beyond traditional PDF readers with 5 built-in, 100% offline document creation & editing studios:

- **Word Studio (.docx, .doc):** Rich-text word processor with heading presets, custom typography, alignments, image inserts, and instant Word/PDF export.
- **Spreadsheet Studio (.xlsx, .xls, .csv):** High-performance 2D formula grid supporting `=SUM`, `=AVERAGE`, `=COUNT`, cell styling, auto-formatting, and multi-sheet workbooks.
- **Presentation Studio (.pptx, .ppt):** 16:9 interactive slide deck designer with animated thumbnail strip, layout templates, color themes, and full-screen presentation mode.
- **HTML Web Studio (.html, .htm):** Triple-view HTML/CSS editor with live real-time split-screen DOM rendering, viewport switcher (Mobile/Tablet/Desktop), and auto-formatter.
- **Universal Code & Text IDE:** High-speed code editor for 50+ programming languages (JSON, Dart, JS, Python, SQL, XML, TXT) with search & replace and line counters.

3. Zero-Latency High-Speed Images to PDF

Optimized image-to-PDF compilation pipeline eliminates CPU bottle-necks:

- **Direct Byte Pass-Through:** When images are in standard orientation without post-filters, raw byte streams are embedded directly into PDF raster pages, creating multi-page PDFs instantly.

- **Optional Enhancement Filters:** Clean Scan, Document Magic Color, and B&W; high-contrast modes available for specialized document scanning.
- **Direct .gdrm Export:** One-tap option to seal photo scans directly into zero-trust encrypted .gdrm files without intermediary saving.

4. Universal 'Open With' Association & Back-to-App Flow

Seamless operating system integration on Android and Windows:

- **Universal File Associations:** Zen PDF is registered as the default handler for .pdf, .gdrm, .docx, .pptx, .xlsx, .csv, .html, .json, and .txt.
- **Seamless Back-to-App Exit Flow:** When documents are opened from WhatsApp, Gmail, Telegram, or file managers, closing the viewer immediately invokes SystemNavigator.pop() to return directly to the calling app without trapping the user on Zen PDF's home screen.

5. Digital Signature Verification & Stamping (Aadhaar & PAN)

- **Aadhaar & PAN Certificate Verification:** Analyzes embedded SHA-256 PKCS#7 digital signatures, transforming unverified yellow question marks (?) into authentic Adobe Acrobat green ticks (✓).
- **Digital Signing:** Stamp verified digital certificates with signer name, location, timestamp, and custom approval reasons.
- **Whiteout & Annotations:** Cover sensitive numbers or add ink signatures with pixel precision.
- **Page Organizer:** Drag-and-drop page reordering, merging, rotating, and splitting.

6. Feature Specifications & Compatibility Matrix

Component	Key Capabilities	Supported Formats
GDRM Security	Hardware Lock (\$K_{device}\$), ChronoLock, Watermark, SnailTrail	.gdrm, .pdf
Universal Studio	Word, Spreadsheet, Presentation, HTML, Code IDE	.docx, .xlsx, .pptx, .html, .json, .txt
Chrome PDF Viewer	120 FPS render, search, infinite zoom, rotate, thumbnails	.pdf, .gdrm
Fast Image Scanner	Zero-latency pass-through, multi-page binding	.jpg, .png, .webp, .heic
Signature Engine	PKCS#7 SHA-256 verification & Adobe Acrobat checkmark	Aadhaar, PAN, Invoices, Contracts
OS Integration	System 'Open With' intent + Instant Back-to-App pop	Android 7.0+ / Windows 10/11

7. Privacy & Offline-First Security Guarantee

Zero Cloud Dependency • 100% On-Device Privacy:

Zen PDF Studio and the GDRM SDK operate entirely on-device. No telemetry, analytics, ad networks, or cloud trackers are bundled. All cryptographic hashing, document rendering, and PDF conversions occur within isolated local application memory.